

to tune the noise performance, signal bandwidth, or latency.

As shown in Table 2, a higher over-sampling ratio (OSR) implementation leads to better accuracy and higher sample resolution but less signal bandwidth and higher latency. Similarly, lowering OSR reduces accuracy and sample resolution but increases bandwidth and reduces latency.

A similar DSP or an FPGA can also implement such a digital filter.

Additionally, as shown in Fig. 3, multiple digital filters can be implemented in parallel to achieve higher sample resolution, lower latency, and higher bandwidth, all at the same time. One of the digital filters can implement a high OSR digital filter for better noise performance, and another one can implement a low latency digital filter.

With the system advantages offered by an isolated modulator-based solution, there is a resulting trend to move to an isolated modulator-based solution in high-performance systems.

Isolated modulators in traction inverters

Fig. 4 shows an implementation of an isolated modulator-based solution in automotive traction inverters.

Traction inverters have a direct influence on the driving experience and require precise control of speed and torque of the traction motor. A shunt coupled with an isolated delta-sigma modulator provides the highest-quality feedback signals to the controller in order to establish the pulse-width modulation (PWM) pattern for bridge transistors. The digital filter implementation allows the engi-

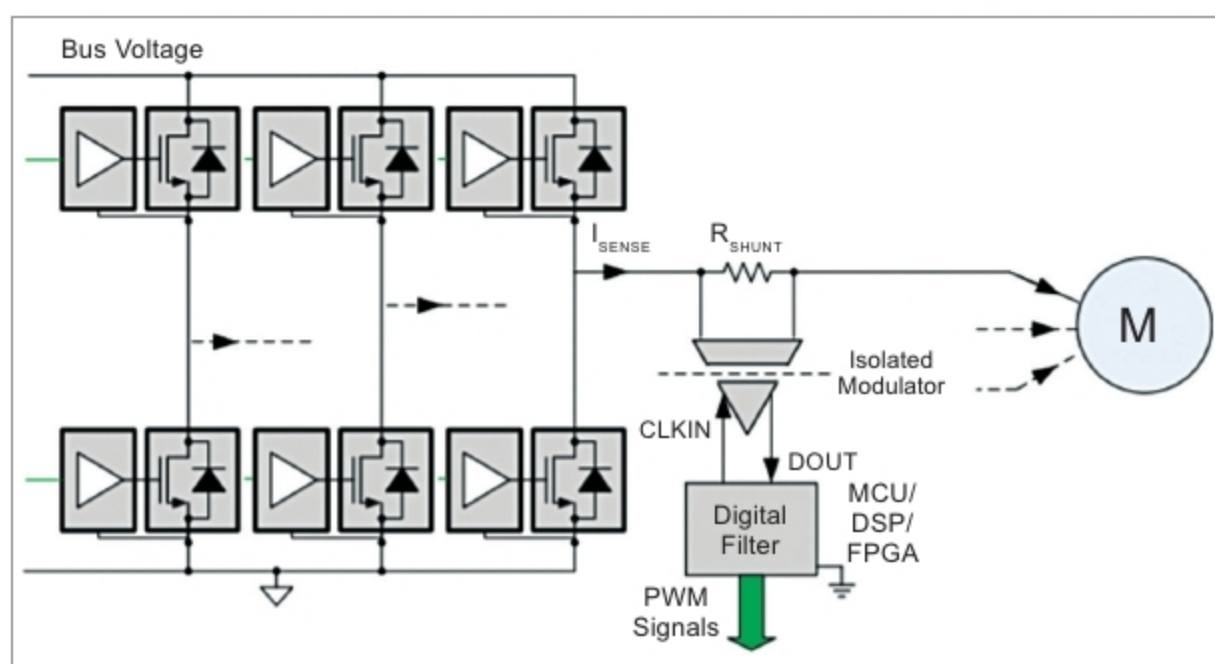


Fig. 4: Current measurement using an isolated modulator

Recommended Devices

Table 3

Device	Isolation	Description
AMC1306	Reinforced	± 50 -mV, ± 250 -mV small isolated modulators
AMC1305, AMC1305-Q1	Reinforced	± 50 -mV, ± 250 -mV isolated modulators
AMC1301, AMC1301-Q1	Reinforced	± 250 -mV isolated amplifiers
AMC1302, AMC1302-Q1	Reinforced	± 50 -mV isolated amplifiers
AMC1311, AMC1311-Q1	Reinforced	0-V to 2-V isolated amplifiers

neer to tune the quality of traction motor controls.

As shown in Fig. 3, FPGA, MCU, and DSP can have multiple digital filters running in parallel. One of the digital filters can be a high-performance digital filter that provides accurate feedback signals to control the bridge transistors. Another digital filter can be a low-latency digital filter for detecting overload or overcurrent conditions. A third-order (Sinc³) filter with a different OSR can be used for both digital filters.

Table 3 lists the recommended devices for use with the isolated amplifiers and modulators.

Conclusion

Isolated modulators offer higher sample resolution and accuracy compared to isolated amplifiers. With the combination of isolated modulators and custom digital filters, engineers can trade system latency and bandwidth with sample resolution. Isolated modulator-based solutions require fewer components and enable a smaller solution size at a reasonable cost. Isolated modulators are strongly recommended in isolated measurement applications wherein high sample resolution or low latency is required. **EFY**

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